



PROJECT TITLE PROPOSAL

1.0 Name of Proponents:

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2.0 Proposed Case Study Title

Integrated Hotel Management System with Reservation, Revenue,
Monitoring, Guest Services, and Parking Management

3.0 Area of Investigation

The study will be conducted at East Asia Royale Hotel, National
Highway, Barangay Lagao, General Santos City, South Cotabato 9500,
Philippines.,

4.0 Cited Problems

This study aims to address the existing problems encountered by hotel
management in handling parking operations, reservations, guest services,
housekeeping, billing, monitoring, and financial management using
traditional manual processes.

4.1 The hotel has manually recording, tracking and assigning available
parking slots, making it difficult for staff to identify vacant spaces. This



results in inefficient parking allocation, overcrowding, and disorganized parking operations.

4.2 The hotel relies on manual reservation and guest registration processes, increasing the risk of double bookings, inaccurate room assignments, scheduling conflicts, and delays during guest check-in and check-out.

4.3 The hotel has no reliable system for recording guest vehicle arrivals and departures, making it difficult to monitor vehicles within the premises and maintain accurate parking records.

4.4 The use of handwritten or paper-based entry slips for guests and vehicles may result in lost, misplaced, or unauthorized records, causing delays in verification at the gate and front desk.

4.5 Manually recording the housekeeping management makes it difficult for housekeeping staff and front desk personnel to track room cleaning status, maintenance updates, and room availability, affecting room turnover efficiency.

4.6 Manual computation of room charges, additional services, and parking fees increases the possibility of calculation errors, inconsistent billing, and disputes during guest checkout.



4.7 The hotel lacks a structured method for collecting guest ratings and comments, making it difficult for management to evaluate service quality and identify areas that require improvement.

4.8 Hotel management lacks a centralized view of important operational data, such as occupancy levels, parking usage, and revenue performance, making real-time monitoring and decision-making more challenging.

4.9 Hotel staff have limited ability to track and analyze income generated from room bookings, additional services, and parking fees, affecting financial performance evaluation.

4.10 The absence of automated occupancy reports makes it difficult for management to analyze room demand patterns and properly plan staffing and resource allocation.

4.11 The lack of automated daily, weekly, and monthly financial reports makes it difficult for hotel management to evaluate overall business performance and support effective financial planning.

5.0 Functions of the Proposed System

The proposed Integrated Hotel Management System with Reservation, Revenue, Monitoring, Guest Services, and Parking Management functions as a digital platform that assists hotel management in handling guest reservations, room monitoring, billing, housekeeping, and parking transactions. The system provides tools for recording guest



bookings, validating reservations, managing check-in and check-out, tracking parking slot usage, monitoring revenue, and generating reports that help improve accuracy, reduce manual errors, and support better hotel management decisions.

6.0 Features of the Proposed System

Parking Slot Management

Enables the hotel to organize, allocate, and monitor available parking slots in real time, allowing staff to efficiently assign parking spaces to arriving guests and prevent overbooking of parking areas.

Reservation Management, Validation, and Guest Check-in/Check-out

Provides hotel staff with a centralized system for managing room and parking reservations, allowing them to create, modify, confirm, or cancel bookings while monitoring room and parking slot availability across different dates. The system automatically validates reservations to prevent double bookings and scheduling conflicts and streamlines the guest check-in and check-out process by digitally managing room assignments, guest information, and billing records, reducing front desk delays and improving overall operational efficiency.

Vehicle Check-in and Check-out

Records the arrival and departure of guest vehicles, capturing essential details such as plate number, time of entry, and time of exit to maintain accurate parking activity logs.



Digital Entry Pass

Generates a digital pass containing reservation and vehicle details that guests can present at the gate or front desk, replacing manual entry slips and reducing processing time during arrival.

Housekeeping Management

Coordinates room cleaning, maintenance requests, and room status updates between housekeeping staff and the front desk to ensure rooms are prepared and available for incoming guests.

Billing Management

Automatically computes room charges, additional services, and parking fees into a consolidated guest bill, reducing manual computation errors and ensuring accurate, transparent charges.

Guest Feedback

Collects guest ratings and comments regarding their stay, enabling hotel management to identify service strengths and areas for improvement based on direct guest input.

Monitoring and Analytics Dashboard

Presents a consolidated visual overview of hotel operations, including occupancy levels, parking usage, and revenue performance, to support real-time decision-making by hotel management.

Revenue Monitoring

Tracks income generated from room bookings, additional services, and parking fees, allowing management to evaluate financial performance



across different periods.

Occupancy Reports

Generates detailed reports on room occupancy rates, helping management assess demand patterns and plan staffing and resource allocation accordingly.

Financial Reports

Produces summarized daily, weekly, and monthly financial reports covering room revenue, service charges, and parking income to support accurate business evaluation and planning.

7.0 Programming Language and Database to be used:

Front End: HTML, CSS, and JavaScript

Back End: PHP, Framework Laravel

Database: MySQL Database

8.0 Software Engineering Model to be Used



The proposed Integrated Hotel Management System with Reservation, Revenue, Monitoring, Guest Services, and Parking



Management will use the Agile Software Development Model. This model is suitable for the study as it supports incremental development, continuous testing, and improvements based on user feedback. It ensures that the system features, such as reservation management, parking slot monitoring, guest check-in and check-out, and report generation, meet the needs of hotel management while maintaining functionality, usability, and reliability.

9.0 Reasons for the Choice

The development of this system is motivated by the following goals:

- 9.1** To establish a digital parking slot management system that will allow hotel staff to organize, allocate, and monitor available parking spaces in real time, reducing overcrowding, parking conflicts, and inefficient use of parking areas.
- 9.2** To develop a centralized reservation management, validation, and guest check-in/check-out system that will allow hotel staff to manage room and parking reservations, verify booking details, prevent double bookings, and efficiently process guest arrivals and departures.
- 9.3** To provide a vehicle check-in and check-out recording feature that will track guest vehicle movement, record entry and exit details, and maintain accurate parking activity logs.
- 9.4** To introduce a digital entry pass system that will replace manual entry slips, allowing faster and more reliable verification of guest and vehicle information at the gate and front desk.
- 9.5** To establish a housekeeping management system that will



coordinate room cleaning schedules, maintenance updates, and room status information between housekeeping staff and the front desk for faster room preparation and turnover.

9.6 To implement a billing management system that will automatically calculate room charges, additional service fees, and parking fees, ensuring accurate, transparent, and efficient guest billing.

9.7 To integrate a guest feedback feature that will allow hotel management to collect guest ratings and comments, helping identify service strengths and areas that require improvement.

9.8 To develop a monitoring and analytics dashboard that will provide a consolidated view of hotel operations, including occupancy levels, parking usage, and revenue performance to support effective decision-making.

9.9 To provide a revenue monitoring system that will track income generated from room bookings, additional services, and parking fees, allowing management to evaluate financial performance.

9.10 To generate occupancy reports that will help hotel management analyze room demand patterns, monitor occupancy rates, and plan staffing and resource allocation effectively.

9.11 To provide financial reports that will summarize daily, weekly, and monthly business performance, including room revenue, service charges, and parking income, supporting accurate financial evaluation and planning.



10.0 Importance of Study

The Integrated Hotel Management System with Reservation, Revenue, Monitoring, Guest Services, and Parking Management is important for the following reasons:

Improved Reservation Record Management

The system provides a digital reservation record that allows hotel staff to easily record, monitor, and manage guest bookings. This reduces errors, prevents lost records, and makes reservation tracking more efficient.

Accurate Room and Reservation Tracking

The system automatically validates reservations and manages room assignments, reducing errors caused by manual scheduling. It helps hotel staff maintain accurate information regarding room availability and guest bookings.

Efficient Parking Slot Monitoring

The system allows hotel staff to manage parking slot availability, vehicle entry and exit records, and allocation details. It helps prevent overcrowding and enables better monitoring of parking usage.

Better Revenue Monitoring and Reporting

The system records daily transactions and generates financial reports, allowing hotel management to analyze room revenue, service charges, and parking income for better decision-making.

Improved Hotel Management Efficiency



By replacing traditional handwritten records with a digital system, the proposed system reduces manual workload and provides faster access to important hotel information, improving the efficiency of daily operations.

Enhanced Guest Experience and Decision-Making

The system provides reports such as guest feedback summaries and occupancy trends, helping hotel management identify service issues and make informed decisions for hotel improvement.

11.0 Target Users/ Beneficiaries

List all target users or beneficiaries of the proposed system.

- Hotel Owner
- Hotel Manager
- Front Desk Staff / Receptionists
- Housekeeping Staff
- Parking Attendants
- Guests
- Developers